

heavily on the oracle exchange to manipulate the price, then proceed to purchase far more on the primary exchange to capitalize on the price movement. The net effect is that the attacker is able to manufacture a discounted price on a high liquidity exchange by manipulating a low liquidity oracle.

Economic exploits become even trickier when considering that flash loans allow any Ethereum user to become financially equipped for a single transaction. Special care must be used when designing protocols such that they cannot be manipulated by massive market volatility within a single transaction. An economic exploit that uses a flash loan is a *flash attack*. A series of high-profile flash attacks were executed in February 2020 on bZx Fulcrum, a lending market similar to Compound.<sup>4</sup> With a flash loan, the attacker diverted some of the funds to purchase a levered short position and used the rest to manipulate the price of the oracle exchange on which the short position was based. The attacker then closed the short at a profit, unwound the market trade, and paid back the flash loan. The net profit was almost \$300,000 worth of funds previously held by bZx, for near zero up-front cost.

The most famous smart contract attack occurred in 2016. Designed by Slock.it to act as the first decentralized venture capital fund for blockchain ventures, it was launched April 30, 2016,<sup>5</sup> and attracted about 14 percent of all the ether available at the time. The DAO tokens began trading in May 2016, but a crucial part of the code had two lines in the wrong order, allowing ether to be repeatedly withdrawn – before